MILITARY SPECIFICATION

PUMPS, RECIPROCATING, POWER-DRIVEN:

DIAPHRAGM, GASOLINE-ENGINE-DRIVEN,

WHEEL-MOUNTED

This interim amendment is approved for use within the US Army Belvoir Research, Development and Engineering Center, Department of the Army, with MIL-P-509F, dated 7 November 1983.

PAGE 2

2.1.1, under "SPECIFICATIONS, MILITARY", add:

"MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package."

PAGE 3

2.1.1, under "STANDARDS, MILITARY", delete MIL-STD-1188 in its entirety and add:

"MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities."

PAGE 6

3.6, delete in its entirety and substitute the following:

"3.6 Noise limits. The noise produced by the pump unit shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.4. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474,

AMSC N/A FSC 4320

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4.3, except the sign will state "HEARING PROTECTING REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign."

PAGE 8

3.15.1, at the end of the paragraph, add the following new sentence:

"The engine shall be selected in accordance with MIL-STD-1410 for a Class II end item."

PAGE 9

3.15.1.2, line 3, after sentence ending "...feed systems.", add the following new sentence:

"A label stating "MOGAS ONLY" shall be permanently attached on or near the fuel fill port in accordance with MIL-STD-1472, 5.5.1.1 (see 3.25)."

PAGE 11

3.19, at the end of the paragraph, add the following new sentence:

"The tie-down provisions shall be labeled "TIEDOWN HERE" in letters 0.75 inches high of a color contrasting with the background color in accordance with MIL-STD-1472, 5.9.11.3.9 (see 3.25)."

PAGE 13

After 3.24.6, add the following new paragraph:

"3.25 Human factors engineering. The pump shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), audio displays (5.3), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable."

After 4.1.1, add the following new paragraph:

"4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted

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to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 14

4.5.1, add the following new examination:

"113a. Human factors engineering not as specified. "

PAGE 16

Delete 4.5.2.4 thru 4.5.2.4.3 in their entirety and substitute the following new paragraph:

"4.5.2.4 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test."

PAGE 18

4.5.2.10, lines 7 thru 10, delete last sentence and substitute the following:

"These readings shall be compared with the engine speed (rpm), brake horsepower, and intake manifold vacuum curve for the continuous brake horsepower condition as established during performance of Test Method Series 2000 of MIL-STD-1400. Any intake manifold vacuum reading recorded during the performance test (see 4.6.2.4) that is lower than the reading listed on the continuous horsepower curve established during performance of MIL-STD-1400, Test Method 2300, at the corresponding speed (rpm), shall constitute failure of this test."

PAGE 19

4.6.2, delete "pack" and substitute "packaging".

4.6.2.3, line 2, delete "2.5" and substitute "1.0".

Column 3, delete heading "Comm" and substitute "C".

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Defect # 114, column 3, add "5.2.3".

Defect # 115, column 3, add "5.2.3".

Defect # 118, column 3, delete "5.2.3" and substitute "5.2.3.2".

Defect # 119, column 3, add "5.2.3".

Defect # 123, column 3, add: "5.2.3"

Defect # 124, column 3, add: "5.2.3"

PAGE 20

4.6.2.3, column 3, delete heading "Comm" and substitute "C".

Defect # 125, column 3, add "5.2.3".

Defect # 127, column 3, add "5.3.3".

Defect # 130, column 3, delete "5.4.2" and substitute "5.4".

5.1, lines 2 and 6, delete "preproduction" and substitute "first article".

5.2, delete "commercial" and substitute "C".

PAGE 21

5.2.1.5, line 2, add at the end of the sentence:

"except the coolant system shall be filled with premixed equal parts (by volume) of antifreeze (ethyl glycol) conforming to MIL-A-46153 and clean water."

PAGE 22

5.2.2.1, line 2, delete "Domestic" and substitute "domestic".

5.2.3, delete in its entirety and substitute the following:

"5.2.3 Level C. Each complete pump unit with loose components, provisioning items, hoses (when required), exterior surfaces and maintenance tools shall be preserved as specified for level B, except as otherwise specified herein (see 5.2.3.1 and 5.2.3.2)."

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After 5.2.3.1, add the following new paragraph:

"5.2.3.2 Pumps. The interior surfaces of the pump casings and all parts within the casings shall be coated with an appropriate preservative lubricating oil. The suction and discharge openings shall be sealed with plastic or metal caps, plugs or blank flanges."

5.3, line 1, delete "commercial" and substitute "C".

5.3.1, line 1, delete "5.2.1" and substitute "5.2".

5.3.2, line 1, delete "5.2.2" and substitute "5.2".

5.3.3, delete in its entirety and substitute the following new paragraph:

"5.3.3 Level C. Each complete pump unit with the consolidated items shall be packed in wooden box that will prevent damage to the contents during shipment. The appropriate blocking and bracing material shall be used to prevent damage to the pump and consolidated items."

PAGE 23

Delete 5.4 thru 5.4.2 in their entirety and substitute the following new paragraph:

"5.4 Marking. In addition to any special or other identification marking required by the contract or order (see 6.2), consolidated containers and shipping containers for each pump unit packed as specified in 5.3.1, 5.3.2 and 5.3.3, shall be marked in accordance with MIL-STD-129."

PAGE 24

6.2, line m, delete "preproduction" and substitute "first article".

Line n, delete "preproduction" and substitute "first article".

Line o, delete in its entirety and substitute:

"o. Level of preservation and packing required (see 5.2 and 5.3)".

Line q, delete "5.4.1" and substitute "5.4".

Custodian:
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Preparing activity:
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